Competition and Pricing Dynamics in the UK Espresso Machine Digital Market

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Data Preparation

1.1 Data Volume

The empirical foundation utilises a primary dataset extracted from Amazon.co.uk on the 7th March 2026. Targeting the espresso machine category simulated a representative consumer search journey. A VBA script, provided in Appendix 7.3, initially acquired 309 unique observations identified by Amazon Standard Identification Numbers (ASIN), including both organic and algorithmically injected sponsored advertisements. Scraping all seven result pages captured the active digital shelf, ensuring no significant brands or variants were omitted. Filtering removed duplicates and unpriced observations to focus on organic competition. This resulted in a refined baseline of 303 observations for the market universe.

1.2 Data Breadth

To facilitate a comprehensive competition analysis, the dataset includes product characteristics beyond pricing. Using an invoked function within Power Query, the dataset was expanded to include brand, pump pressure, capacity, weight, review scores, review count, and special features. Technical specifications extracted from these features were used to create dummy variables regarding the presence of integrated grinders, milk-frothing systems, and manual operation. This breadth of data allows for the quantification of quality dimensions and technological features that directly influence consumer willingness to pay. Following the automated scrape and the removal of duplicates, the dataset comprised 303 observations, which provided a comprehensive foundation for the cleaning and enrichment phase.

1.3 Data Cleaning

The raw metadata underwent transformation to ensure suitability for econometric modelling. Observations lacking brand names or prices were removed to preserve integrity. The pump pressure variable suffered from near-total missingness, which necessitated its removal. Capacity and weight specifications were cleaned by stripping non-numeric strings and standardising values into litres and kilograms, respectively. A manual outlier audit corrected unit anomalies and refined the market definition by removing non-espresso machines. To address a 30% completion rate for weight, a significance-weighted enrichment strategy used manual research to recover data for market-leading brands. This yielded 183 observations for concentration analysis and a sub-sample of 111 observations for controlled regression analysis (representative sample provided in Appendix 7.1).

1.4 Methodology

The data was assembled using a VBA script and Power Query to scrape metadata directly from the Amazon HTML structure. The primary challenge involved managing the high computational demand against local hardware constraints. Recursive server calls caused significant system instability and frequent software crashes. To resolve this, the data ingestion process was manually controlled by periodically pausing the connection to the Amazon server. This ensured stable migration of captured metadata from the Power Query cache directly into the Excel worksheet. Breaking collection into manageable stages preserved the raw dataset's integrity while preventing further system failures.

Market Overview

2.1 Product Description

Espresso machines extract coffee by forcing pressurised water through grounds to produce a rich crema, serving as the base for beverages such as lattes and cappuccinos. The UK market ranges from entry-level manual machines to high-end automated bean-to-cup systems, characterised by high brand differentiation and technical complexity. Manufacturers compete on quality dimensions, including extraction consistency and integrated features. As consumers prioritise brand reputation and quality alongside technical specifications, established brands like De'Longhi and Sage maintain premium pricing, while newer entrants often compete on aesthetic appeal.

2.2 Channel Availability

Identified products are available via major omnichannel retailers such as Currys, John Lewis, and Argos, which seamlessly integrate digital platforms with physical showrooms. These brands maintain a dual-presence model, enabling consumers to inspect build quality in-store before selecting home delivery or immediate purchase. This hybrid approach ensures accessibility for both convenience-oriented online shoppers and traditional retail consumers, effectively bridging the gap between digital and physical marketplaces.

2.3 Price Comparison

A comparison of models reveals that pricing is largely consistent across channels with minor variations driven by platform-specific promotions. For instance, the De'Longhi Dedica Arte is £250 at Argos compared to £217 on Amazon, while the Sage Barista Express ranges from £499.95 at John Lewis to £529.95 at Argos. While physical stores face higher overheads like rent and staffing, that often result in a price premium, the total cost of online purchases can be equalised when factoring in delivery fees for the consumer.

2.4 Platform Competition

Online platforms increasingly engage in non-price competition, differentiating through value-added services rather than base price alone. While core product listings overlap, Amazon and John Lewis compete via delivery logistics and extended warranties. Amazon utilises its extensive repository of user reviews to mitigate search frictions, whereas specialised retailers like John Lewis prioritise after-sales support and premium brand positioning. Furthermore, algorithmic prominence, determined by search rankings and display order, acts as a critical driver of consumer visibility and competitive pressure within the digital marketplace.

2.5 Cross-Channel Competition

Clear evidence of cross-channel competition exists as stores often match online prices to prevent showrooming. Product offerings are remarkably similar across both channels although physical stores may stock a more curated selection of high-demand models. The primary difference remains immediate availability and the ability to test products in physical retail versus the extensive variety and delivery convenience of online platforms.

2.6 AI Versus Market Data

An inquiry to Gemini regarding market leading espresso machine brands identified De'Longhi, Sage, Gaggia, Krups, Smeg, Nespresso, Jura, Philips, Melitta, and Breville. This mirrors the report's findings regarding De'Longhi and Nespresso, but Gemini exhibited a reputational bias toward premium legacy brands like Gaggia and Jura. These brands showed lower listing volumes in the live dataset compared to budget-focused entrants like Wacaco. Such discrepancies suggest that AI outputs may prioritise historical brand authority over real-time market trends or regional UK promotional shifts, reflecting the lag between static training data and live marketplace dynamics.

Market Concentration

3.1 Manufacturers

The espresso machine market on Amazon UK comprises established global multinationals and numerous smaller generic entrants. Primary players identified in the dataset include De'Longhi, Nespresso, and Krups, which are prominent multinational corporations with extensive manufacturing networks. De'Longhi operates as a major Italian group while Nespresso and Krups are part of the Swiss-based Nestlé and French-based Groupe SEB, respectively. These firms leverage significant economies of scale and brand equity to maintain their positions. The market also includes specialised firms such as Sage which is part of the Australian Breville Group. While these multinationals dominate premium segments, the digital shelf is populated by generic or domestic firms focusing on entry level machines. This suggests that while high end engineering requires multinational capital, the barriers to entry for basic pump machines are low enough to allow smaller firms to compete on price.

3.2 Concentration Analysis

Market shares are calculated using product listings to represent variety and consumer reviews to proxy for sales volume. These measures assume functional substitutability within the defined market and that manufacturers maintain relatively symmetric cost structures for comparable technology. The competitive landscape is quantified through the Four Firm Concentration Ratio (C4) and the Herfindahl Hirschman Index (HHI).

The C4 ratio is defined as the sum of the market shares of the four largest firms in the industry. This ratio approaches 0% in a perfectly competitive market and reaches a maximum of 100% in a market entirely controlled by four entities.

HHI is defined by the equation,

$$HHI = \sum_{i=1}^{N} (S_i)^2,$$

where the percentage market share S_i of each individual firm is squared and summed for all N firms in the market.

This index ranges from a theoretical minimum of $\frac{10{,}000}{N}$ to a maximum value of 10,000. With 183 firms identified in the current market the theoretical minimum HHI is 54.64.

Table 1: Market Concentration Comparison

Ratio	Count of product listings	Count of reviews
C4 Ratio	18.58%	70.20%
HHI	197.38	1785.54

Results in Table 1 demonstrate that the count of product listings indicates a highly competitive environment, whereas the count of reviews reveals a significantly higher degree of concentration. This disparity suggests that although hundreds of brands are visible to consumers, the actual purchasing activity is concentrated within a small elite group of trusted manufacturers.

3.3 Merger Analysis

The concentration of the market is further quantified using two distinct hypothetical merger scenarios between the firm with the highest market share and the firm with the third-largest share. For the product listing measure this merger was analysed between De'Longhi and Krups. For the reviews measure the merger was analysed between De'Longhi and Nespresso.

Table 2: Impact of Hypothetical Mergers on concentration measures

Ratio	Product Listings with merger between De'Longhi and Krups	Reviews with merger between De'Longhi and Nespresso
C4 Ratio	21.86%	75.28%
HHI	251.72	2382.99

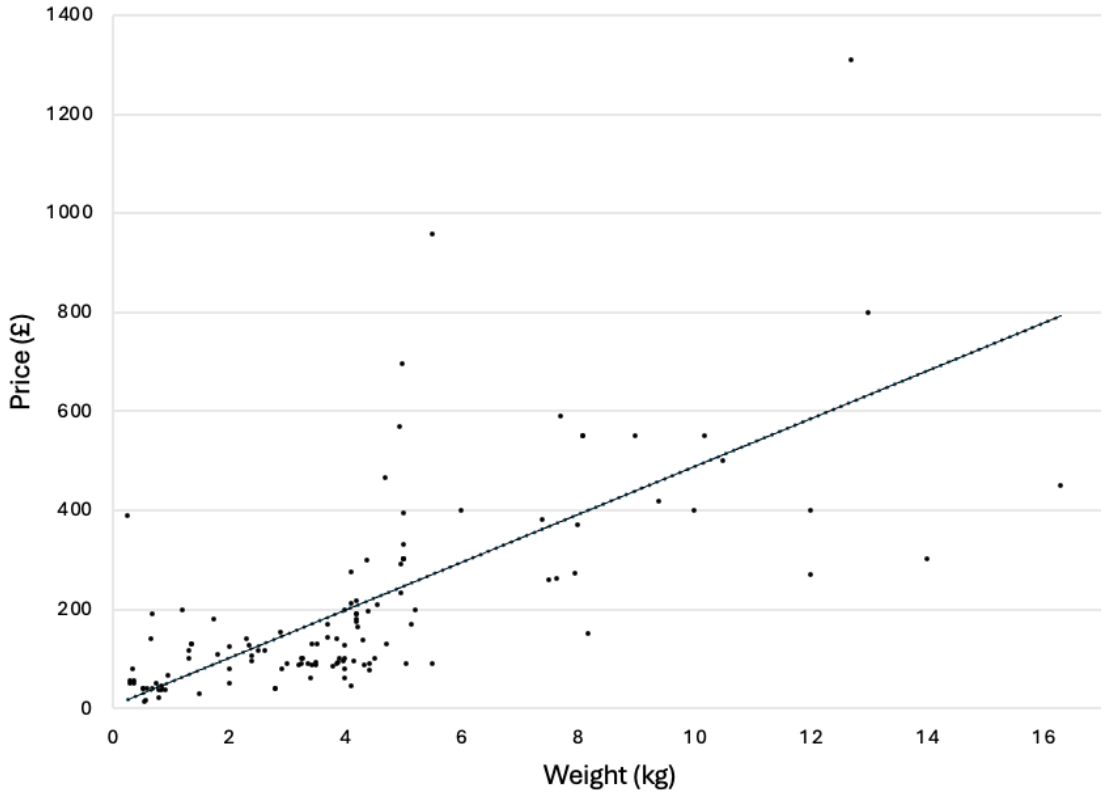
The results in Table 2 show that the merger between De'Longhi and Nespresso would cause the reviews HHI to increase by 597.45 to reach a total of 2382.99. This represents a significant shift that moves the market structure closer to the threshold of high concentration. Such an increase would likely trigger regulatory scrutiny from the Competition and Markets Authority because it represents a substantial reduction in competitive pressure within a moderately concentrated market. This merger would grant the combined entity an overwhelming level of market engagement which could lead to higher prices and a reduction in innovation across the UK espresso machine segment.

Empirical Analysis

4.1 Preliminary Analysis

To establish the empirical foundation for the hedonic models, a bivariate analysis was conducted to explore the relationship between weight and market value. Figure 1 illustrates the correlation, providing a visual justification for including weight as a primary independent variable.

Figure 1: Scatter Plot Illustrating the Correlation Between Weight and Price for Espresso Machines



4.2 Empirical Results

Following the preliminary visual analysis, regression specifications were estimated. Table 3 presents the comparative results.

Table 3: Comparative Hedonic Price Analysis

Model	Naïve (weight)	Naïve (manual)	Linear	Non-linear
Dependent variable	Price (£)	Price (£)	Price (£)	Log of Price (£)
Variables				
Weight (kg)	48.33***		55.99***	0.138***
	(8.664)		(14.06)	(0.0345)
Manual		197.0	196.6	0.257
		(145.6)	(157.0)	(0.212)
Count of reviews			-0.00167	-0.00000640
			(0.00128)	(0.00000561)
Capacity (litres)			-84.24	-0.0149
			(78.11)	(0.199)
Reviews (out of 5)			-53.78	-0.0514
			(48.58)	(0.0787)
Milk Frother			31.07	0.191
			(59.21)	(0.193)
Grinder			2.809	-0.180
			(58.71)	(0.234)
Brand	Not Included	Not Included	Included	Included
Constant	6.123	186.0***	210.8	4.291***
	(27.64)	(14.87)	(243.9)	(0.454)
R^2	0.464	0.041	0.888	0.929
Observations	125	183	111	111

Robust standard errors reported in parentheses.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Full variable definitions are provided in Appendix 7.2.

4.3 Baseline Specification

The analysis begins with two naïve specifications to establish baseline correlations between price and individual product characteristics. As presented in Table 3, an initial naïve regression focusing solely on machine weight explains 46.4 per cent of price variation. A secondary baseline model examines manual functionality, which shows a significantly weaker initial correlation of 4.1 per cent. These specifications provide a necessary contrast to the full hedonic pricing model, which posits that a product is a bundle of characteristics and its price is the sum of the valuations of those individual attributes.

The linear specification can be defined as,

$$P_i = \beta_0 Weight_i + \beta_1 Manual_i + X_i'\beta + \epsilon_i.$$

In this model, P_i represents the price for product i and $X_i'\beta$ is a vector of control variables that incorporates technical features such as capacity, grinder presence, and milk frothing capabilities, alongside categorical brand identifiers. The composite error term ϵ_i contains the constant intercept and the idiosyncratic error term.

Consequently, the full linear specification is justified as it moves the variables within the $X_i'\beta$ vector out of the error term and into the regression model. This addresses the omitted variable bias present in the baseline models, where the exclusion of brand equity and technical specifications caused the regressors to be correlated with the error term. This consolidated approach isolates the impact of tangible engineering from manufacturer prestige, successfully increasing the explanatory power to 88.8 per cent.

4.4 Diagnostics

The full linear regression suggests that weight is the most robust factor of price. Ceteris paribus, each additional kilogram of weight adds £55.99 to the predicted retail price ($p < 0.01$). This reflects a high willingness to pay for build quality. Interestingly, while manual functionality and milk frothers carry positive coefficients, they are not statistically significant in this linear specification when controlling for weight and brand.

The diagnostic phase identified significant heteroscedasticity through a Breusch-Pagan test ($p = 0.000$), confirming that error variance was not constant across the price range. This was addressed by using robust standard errors for all estimates to ensure the reliability of t-statistics. Additionally, a Ramsey RESET test was conducted ($p = 0.000$), indicating that the linear functional form was misspecified.

4.5 Non-Linear Specification

Following the diagnostic evidence from the RESET test, the model was re-estimated using the natural logarithm of price as the dependent variable. This non-linear specification captures 92.9% of price variation and allows for the interpretation of coefficients as percentage effects.

In the non-linear model, the coefficient of 0.138 indicates that a one-kilogram increase in weight is associated with an approximately 13.8% increase in price ($p < 0.01$). While Manual products suggest a 25.7% premium, it remains statistically insignificant, reinforcing the finding that in this Amazon-based market, physical build quality and brand identity are more consistent predictors of price.

4.6 Market Context

The regression findings align with the concept of vertical product differentiation, where firms use high quality physical engineering and reputation to distance themselves from competitors in an oligopolistic landscape (Fiorani, 2017). These results are supported by the analysis of the international espresso market, which identifies that technical product quality and brand perception are primary drivers of market value (Fiorani, 2017). The massive explanatory power of the weight variable and brand controls suggests that manufacturers create competitive advantages through build quality, effectively protecting market share against generic competitors (SEB and WMF, 2016).

This statistical evidence is further validated by regulatory decisions and cultural analyses of the coffee sector. The European Commission (2016) investigation into the SEB and WMF merger highlighted a distinct prosumer segment where consumers prioritise professional grade technical specifications for domestic use. This aligns with the cultural shift identified by Morris (2013), where the move toward espresso signifies a desire to replicate the professional coffee house experience at home through specialised equipment. These sources provide a real world justification for the high valuation of both build quality and brand identifiers found in the model.

Furthermore, the necessity of a log linear specification to achieve a superior statistical fit aligns with observations in the Arçelik and Whirlpool (2024) report. The Competition and Markets Authority noted that technical innovations and brand equity often add proportional, percentage-based value to appliances rather than fixed marginal costs, justifying the percentage premium found in the non-linear model. While the market faces constraints from new entrants and private labels, the robust willingness to pay for machine weight suggests that in the premium espresso segment, technical differentiation remains a more potent price determinant than pure cost leadership strategies (SEB and WMF, 2016).

Synthesis and Reflection

5.1 Conclusion

Analysis of the Amazon UK marketplace identifies a vertically differentiated oligopoly characterised by high concentration among multinational incumbents. Structural findings from Section 3 confirm that a limited number of firms maintain significant market power, while empirical results in Section 4 identify physical weight as a statistically significant determinant of price. These results indicate that competition is driven by technical differentiation and brand reputation, as the platform ecosystem reinforces established brand authority.

The structural analysis in Section 3 acknowledges that review volume, while a standard proxy for market power, remains constrained by brand-specific review-to-sale variance. Furthermore, the robustness of the econometric findings is limited by the OLS exogeneity assumption. This requires the error term ϵ_i to be uncorrelated with the regressors X. If this condition fails such that $Cov(X, \epsilon_i) \neq 0$, the resulting coefficients become biased and inconsistent. To mitigate cross-sectional constraints, future research should leverage longitudinal panel data to evaluate the temporal stability of engineering-led price premiums. Firm-level revenue data would permit a decomposition of market power, distinguishing between technical innovation and brand equity to facilitate a more robust competition analysis.

5.2 Lecture Methods

In accordance with the core theoretical frameworks established in Economics of Platforms (Belleflamme and Peitz, 2021), the empirical analysis in this report applies their models of strategic interaction and platform competition. The log-linear transition in Table 3 aligns with hedonic pricing theories where attributes yield proportional utility increments in premium durable goods markets, rather than fixed marginal increases. The evidence of an uncompetitive market structure, which was characterised by high concentration levels where the C4 index reached ≈ 70% based on review volume, aligns with theories regarding market power and the role of reputation as a barrier to entry. These reputational advantages serve as a mechanism to reduce search frictions for established firms, which effectively heightens the information asymmetry faced by newer entrants. Furthermore, the application of hedonic pricing to isolate tangible attributes like weight and capacity mirrors the module's econometric approach to resolving omitted variable bias. By synthesising these results with established theories on entry barriers and ecosystem compatibility, this report demonstrates how technical differentiation and platform reputation interact to dictate competitive outcomes in modern marketplaces.

AI Declaration

6.1 Declaration

I declare the use of Gemini to assist in the production of this report. Gemini was utilised to generate VBA code to ensure a successful output and provided technical proofreading for terminological accuracy and consistency. The tool was also used to extract key details from regulatory documents and academic papers, which were independently sourced for the report.

6.2 Ethical Justification

The use of AI complied with academic integrity guidelines, as the identification and selection of all academic and regulatory sources remained original work. The core market analysis, the execution of the scraping script, and the selection of the final regression models remained original work. The primary dataset was collected and cleaned independently, as evidenced in the provided Appendix.

Technical Appendix

7.1 Data Sample

ASIN	Count of reviews	Brand	Price (£)	Capacity (litres)	Weight (kg)	Reviews (Out of 5)	Milk frother	Grinder	Manual	Special features
B00G5YOVZA	44345	Nespresso	95	0.80	2.4	3.9	0	0	0	Programmable
B06X3Z9MF3	40369	De'Longhi	180	1.00	4.2	4.2	0	0	0	Auto Shut-Off
B06WGTZ874	40369	De'Longhi	165	1.00	4.22	4.2	1	0	0	Milk Frother
B000GXZ2GS	20614	AeroPress	34	0.30		4.7	0	0	1	Auto Clean, Manual, Portable
B06XJ3MFMN	20123	KRUPS	115	0.60	2.5	4	0	0	0	Programmable
B0763N7WMP	13422	Cecotec	130	1.50	1.35	4.1	1	0	0	Steam Wand, Fully Automatic
B08C96BG9H	13033	De'Longhi	137	1.00	4.3	4.2	1	0	0	Milk Frother
B073ZL4Y7F	9663	Nespresso	109	0.60	1.81	4.4	0	0	0	Programmable
B07NXQ7TD5	6310	Godmorn	22	0.45	0.79	4.1	0	0	0	Removable Tank
B01M4J94WY	5694	WACACO	50	0.07	0.29	4.2	0	0	0	Portable
B01MG4VZCT	5570	Nespresso	105		2.4	4.2	0	0	0	Programmable
B01MTZ419O	5337	De'Longhi	195	1.16	4.4	4.6	1	0	0	Milk Frother
B00VTA9F6U	5180	WACACO	50	0.07	0.36	4.4	0	0	1	Manual operation, portable
B0BRKG7X22	4956	OutIn	140	0.08	0.671	4.3	0	0	0	Portable, Auto Shut-off

7.2 Variable Definitions

Variable	Definition	Data Type
Price	Value of the product In pounds	Numeric
Log of price	The natural log of Price	Numeric
Weight	Weight of the product in kilograms	Numeric
Brand	Manufacturer of the product	Categorical
Capacity	Water tank volume of the product	Numeric
Review score	Mean consumer rating	Numeric
Count of Reviews	Total number of user reviews	Numeric
Manual	1 if the machine is manual	Binary
Milk Frother	1 if an integrated milk frothing system is included	Binary
Grinder	1 if an integrated grinder is included	Binary

7.3 VBA Script

```
Sub Final_Espresso_Market_Scraper()
    ' Objects for browser automation and HTML parsing
    Dim IE As Object, doc As Object, elements As Object, element As Object
    Dim i As Integer, row As Integer
    Dim amazonURL As String, asinValue As Variant

    ' Initialise the browser in the background for efficiency
    Set IE = CreateObject("InternetExplorer.Application")
    IE.Visible = False

    row = 2
    Cells(1, 1).Value = "ASIN"
    Cells(1, 2).Value = "Raw Metadata"

    ' Loop through the first 7 pages of the Amazon search results
    For i = 1 To 7
        amazonURL = "https://www.amazon.co.uk/s?k=espresso+machine&page=" & i
        IE.Navigate amazonURL

        ' State-check: Ensure the page is fully rendered before extraction
        Do While IE.Busy Or IE.ReadyState <> 4: DoEvents: Loop

        ' 5-second pause to prevent server-side throttling (Anti-Scraping measure)
        Application.Wait Now + TimeValue("00:00:05")

        Set doc = IE.document
        Set elements = doc.getElementsByTagName("div")

        For Each element In elements
            ' Extracting the unique Amazon Standard Identification Number (ASIN)
            asinValue = element.getAttribute("data-asin")

            ' Validation check to filter out non-product layout containers (Null Validation)
            If Not IsNull(asinValue) Then
                If asinValue <> "" And Len(asinValue) = 10 Then
                    ' Ensures only unique observations enter the dataset
                    If Application.WorksheetFunction.CountIf(Range("A:A"), asinValue) = 0 Then
                        Cells(row, 1).Value = asinValue
                        Cells(row, 2).Value = element.innerText
                        row = row + 1
                    End If
                End If
            End If
        Next element
    Next i

    ' Clean up processes
    IE.Quit
    Set IE = Nothing
    MsgBox "Scrape Successful: " & (row - 2) & " unique observations collected.", vbInformation
End Sub
```

Bibliography

Amazon (2026) *Espresso Machines Marketplace Data* (accessed 7th March 2026).

Arçelik A.Ş. and Whirlpool Corporation (2024) *Anticipated joint venture between Arçelik A.Ş. and Whirlpool Corporation: Final report*. London: Competition and Markets Authority.

Belleflamme, P. and Peitz, M. (2021) *The Economics of Platforms: Concepts and Strategy*. Cambridge: Cambridge University Press.

European Commission (2016) *Case M.8091 - SEB / WMF: Commission decision pursuant to Article 6(1)(b) of Council Regulation (EC) No 139/2004*. Brussels: European Commission.

Fiorani, G. (2017) *The economic analysis of the international espresso coffee market*. Unpublished master's thesis.

Morris, J. (2013) 'Why espresso? Explaining changes in European coffee preferences from a production of culture perspective', *European Review of History: Revue européenne d'histoire*, 20(5), pp. 881-901.